

STEFANO.PELUSSO@UNIMIB.IT

MATERIAL: - CLASS NOTES
- RECORDINGS
- HHWS
- NOTES

TO-DO LIST

- POINT ESTIMATION (POPULATION VS SAMPLE)
- CONFIDENCE INTERVALS
- HYPOTHESIS TESTING
- BAYESIAN STATISTICS

R CODES
R

WRITTEN EXAM WITH
EXERCISES / THEORY

- ON PAPER
 - INDIVIDUAL
 - QUESTIONS ON R
 - CLOSED-BOOK
- Fill GAPS
 CORRECTING
 WRITE FEW LINES

POINT ESTIMATION

STATISTICAL INFERENCE: PROCESS OF COMING TO SOME CONCLUSIONS
ABOUT A POPULATION, BASED ON DATA COLLECTED IN
A SAMPLE.

POPULATION: SET OF ALL STATISTICAL UNITS

- FINITE
 - Ex.: - POP OF HUMAN BEINGS
 - POP OF COMPANIES IN ITALY
- INFINITE
 - SET OF OUTCOMES OF A PHYSICAL EXPERIMENT

SAMPLE: SUBSET OF THE POPULATION
WE OBSERVE

↳ DATASET

	FEATURE 1	FEATURE 2	...
STAT. UNIT 1			
STAT. UNIT 2			
⋮			
STAT. UNIT. m			

STAT. UNIT (M)

n : SAMPLE SIZE (# UNITS IN THE SAMPLE)

N : POPULATION SIZE

⇒ IN STATISTICS: FEATURES ARE TREATED AS RANDOM VARIABLES

X DESCRIBED BY SOME PROB. DISTRIBUTION

WE DO NOT KNOW THIS DISTRIBUTION $\Rightarrow$ WE ASSUME THAT IT

BELONGS TO SOME STATISTICAL MODEL, THAT IS A

FAMILY OF PROBABILITY DISTRIBUTIONS, INDEXED BY A

PARAMETER θ :

PARAMETER

$$\{f(x; \theta), \theta \in \Theta\}$$

SET OF ALL POSSIBLE VALUES
FOR θ (PARAMETER SPACE)

1 — PMF (DISCRETE RVS)
 \ PDF (CONTINUOUS RVS)

MAKING INFERENCE : LEARNING THE MOST "REASONABLE" VALUE OF θ ,
BASED ON THE SAMPLE

Ex: X : # employees in a company

$X \sim \text{Poisson}(\lambda)$, λ unknown $\lambda > 0 \Rightarrow \Theta = (0, \infty)$
 $\uparrow$
 θ in this example

WE COLLECT A SAMPLE OF $n=100$ COMPANIES
STAT. UNIT

$\Rightarrow$ WE WANT TO LEARN λ

EX: X : LENGTH OF A METAL BAR, WE ASSUME $X \sim N(\mu, \sigma^2)$

$$\theta = (\mu, \sigma^2), \quad \mathbb{H} = \{(\mu, \sigma^2) : -\infty < \mu < +\infty, \sigma^2 > 0\}$$

$m = 50$ BARS

~~CAPITAL~~ θ

$m = 50$ BARS

STAT. UNIT

CAPITAL θ

EXPONENTIAL FAMILY (NOT TO BE CONFUSED WITH EXPONENTIAL DISTRIB.)

WE WILL MOSTLY DEAL WITH DISTRIBUTIONS IN THE EXPONENTIAL FAMILY

QUITE LARGE: BERNOULLI, BINOMIAL, POISSON, NORMAL, BETA, GAMMA

$$f(x; \theta) = h(x) c(\theta) \exp \left[\sum_{j=1}^k w_j(\theta) t_j(x) \right] \quad \text{PDF / PMF}$$

AND $\text{supp}(X)$ DOES NOT DEPEND ON θ .

EX (BINOMIAL RV)

$$f(x; p) = \binom{m}{x} p^x (1-p)^{m-x}, \quad x = 0, 1, \dots, m$$
$$= \binom{m}{x} \cdot (1-p)^m \cdot \left(\frac{p}{1-p} \right)^x$$

$$= \binom{m}{x} (1-p)^m \cdot \exp \left\{ \ln \left(\frac{p}{1-p} \right)^x \right\}$$

$$= \underbrace{\binom{m}{x}}_{h(x)} \underbrace{(1-p)^m}_{c(p)} \exp \left\{ \underbrace{x}_{t(x)} \cdot \underbrace{\ln \left(\frac{p}{1-p} \right)}_{w(p)} \right\}$$

✓ 1st condition

$\text{supp}(X) = \{0, 1, 2, \dots, m\}$ DOES NOT DEPEND ON p ✓ 2nd condition

$\Rightarrow$ BINOMIAL DISTR. BELONGS TO EXP. FAMILY

EX (NORMAL DISTR)

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (x-\mu)^2 \right\}, \quad \begin{array}{l} x \in \mathbb{R} \\ \mu \in \mathbb{R} \\ \sigma^2 > 0 \end{array}$$

$\mu \in \mathbb{R}$
 $\sigma^2 > 0$
 $\text{Supp}(X) = \mathbb{R}$ does not depend on μ or on σ ✓

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} \cdot x^2 - \frac{1}{2\sigma^2} \mu^2 + \frac{1}{\sigma^2} x \mu \right\}$$

$$= \underbrace{1}_{h(x)} \cdot \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{\mu^2}{2\sigma^2} \right\}}_{c(\mu, \sigma^2)} \cdot \exp \left\{ \underbrace{\frac{\mu}{\sigma^2}}_{w_1(\mu, \sigma^2)} x - \underbrace{\frac{1}{\sigma^2} \cdot \frac{x^2}{2}}_{t_2(x)} \right\}$$

✓

$\Rightarrow N(\mu, \sigma^2) \in \text{Exp. family}$

NATURAL PARAMETERIZATION : AN EXP. FAMILY DISTRIB. CAN BE
 REPARAMETERIZED AS:

$$f(x; \theta) = h(x) c^*(\eta) \exp \left\{ \sum_{j=1}^k \overbrace{\eta_j}^{w_j \text{ NOT ANYMORE}} t_j(x) \right\} \quad \left(\begin{array}{l} \text{NATURAL} \\ \text{EXPONENTIAL} \\ \text{FAMILY} \end{array} \right)$$

WHERE $\eta = (\eta_1, \eta_2, \dots, \eta_k)$ IS A TRANSFORMATION OF θ ,
 CALLED NATURAL PARAMETER

$\Rightarrow$ THEN WE CAN EXPLOIT

THE RESULT

$$E[t_j(x)] = - \frac{\partial}{\partial \eta_j} \ln c^*(\eta)$$

(PROOF OMITTED)